HealthTrack: A Remote Monitoring System for Home-Based Cardiac Rehabilitation

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Project Overview and Motivation

Cardiac rehabilitation is a structured program of supervised exercise, education, and risk factor management prescribed after a heart attack, coronary stent, bypass surgery, or similar cardiac event. It improves outcomes for patients with heart disease, yet it remains one of the most underused therapies in cardiology. Among 366,103 eligible Medicare beneficiaries, only 24.4% attended even one session, and only 26.9% of those participants completed the full program of 36 sessions (Ritchey et al., 2020). The traditional model requires patients to travel to a hospital gym several times a week for about 12 weeks, a burden that falls hardest on rural, working, and older patients.

HealthTrack is a remote monitoring platform that makes home-based cardiac rehabilitation safe and supervisable. Patients exercise at home while wearing a heart rate band and use a connected blood pressure cuff and pulse oximeter. HealthTrack collects these vital signs and activity data, compares them against each patient's exercise prescription, alerts the rehabilitation team to unsafe exertion or declining progress, and scores each patient's risk of dropout or clinical decline. A joint scientific statement from the American Association of Cardiovascular and Pulmonary Rehabilitation, the American Heart Association, and the American College of Cardiology concluded that home-based programs are a reasonable option for clinically stable low- to moderate-risk patients who cannot attend a center, with clinical outcomes similar to center-based programs at 3 to 12 months (Thomas et al., 2019). The remaining barrier is the absence of tools that give clinicians the same visibility at home that they have in the gym. HealthTrack is designed to close that gap.

Problem Statement

The specific challenge is remote supervision of exercise for cardiac patients. In a center-based program, a nurse or exercise physiologist watches telemetry during every session and intervenes when heart rate exceeds the prescribed ceiling or blood pressure rises abnormally. At home, no one is watching. Programs that offer home-based rehabilitation today typically rely on

phone calls and paper logs, which reveal problems days after they occur and give no early warning that a patient is disengaging.

The problem affects three groups. Patients who cannot reach a center lose access to a therapy proven to improve outcomes. Rehabilitation staff who run home programs carry clinical responsibility without real-time data. Hospitals forgo the outcomes and revenue tied to completed programs. A naive technical fix creates a second problem: alert fatigue. In a study of 12,671 annotated arrhythmia alarms from bedside monitors, 88.8% were false positives (Drew et al., 2014). A system that forwards every abnormal reading would overwhelm a small rehabilitation team. The project therefore requires more than data collection. It requires patient-specific thresholds, pattern detection that recognizes gradual decline, severity-based prioritization, and risk scoring that tells clinicians which patients need attention first.

Stakeholder Analysis

Figure 1 maps 10 stakeholder groups by influence and interest and shows the data and authority flows between them. Five groups require close management. Patients are the source of all data and determine adoption; they need simple devices, low effort, and confidence that someone is watching. Rehabilitation nurses and exercise physiologists are the primary users and need actionable alerts, clear trends, and a manageable workload. The medical director and cardiologists set exercise prescriptions and alert thresholds and carry clinical liability, so their approval governs every safety rule. IT and clinical informatics staff host, secure, and integrate the system and will reject designs they cannot support. Hospital administrators control staffing and budget and need program metrics that justify the investment.

Payers and the compliance and privacy officer hold high influence with lower day-to-day interest and must be kept satisfied: payers through documentation that supports reimbursement, and compliance through safeguards and audit trails aligned with HIPAA. Caregivers, referring physicians, and device vendors are kept informed. Caregivers support adherence, referring physicians drive enrollment and expect progress summaries, and vendors supply the data feeds the system depends on.

Three conflicts are expected. First, physicians favor conservative thresholds for safety, while nurses need fewer alerts to stay responsive. HealthTrack resolves this with physician-owned, patient-specific thresholds, severity tiers that guarantee immediate delivery of critical alerts, and a daily digest for low-priority events, with threshold changes reviewed against alert history. Second, administrators want broad program data, while compliance demands minimum necessary access. Role-based access control and aggregated administrative reports satisfy both. Third, clinicians want frequent readings, while patients resist burdensome monitoring. Continuous capture is limited to exercise sessions with one resting check per day, and adherence data shows whether that balance needs adjustment.

Preliminary System Requirements

The initial requirements matrix, submitted separately, defines 48 requirements, 28 functional and 20 non-functional. Each carries a permanent identifier, a MoSCoW priority, its stakeholder source, an acceptance criterion, a verification method, and the project phase that delivers it. Thirty-four are rated Must.

The functional requirements fall into six groups. Patient and program management stores enrollment, care team assignments, appointments, and an individualized exercise prescription with a heart rate zone and ceiling (FR-01 to FR-04). Data collection ingests vital signs and activity data, rejects physiologically implausible values, and captures patient-reported symptoms and perceived exertion (FR-05 to FR-09). The alert subsystem evaluates readings against patient-specific thresholds, detects patterns such as consecutive missed sessions or a rising resting heart rate, classifies alerts by severity, and escalates unacknowledged critical alerts to the supervising physician (FR-10 to FR-16). A real-time dashboard pushes readings over WebSockets and ranks patients by alert status (FR-17 to FR-19). The risk module predicts dropout and decline, explains contributing factors, and generates recommendations labeled as system-generated suggestions (FR-20 to FR-24). Cross-cutting requirements cover role-based accounts, a documented REST API, and a seeded synthetic data generator so that no real patient data is used (FR-25 to FR-28).

The non-functional requirements set measurable targets. Security requirements include TLS for all traffic, encryption at rest, short-lived token authentication, least-privilege authorization, and an append-only audit log, consistent with the administrative, physical, and technical safeguards of the HIPAA Security Rule (U.S. Department of Health and Human Services, n.d.). Performance targets include critical alert delivery within 5 seconds and a 95th percentile API response time of 300 milliseconds or less. Maintainability targets include 80% unit test coverage and versioned database migrations. The planned stack is FastAPI, SQLAlchemy with PostgreSQL, and a Dash dashboard.

Risk Assessment and Mitigation Strategies

Table 1 rates the principal technical, operational, and ethical risks by likelihood and impact.

Table 1

Project Risk Register With Likelihood and Impact Ratings

Risk	L	I	Score	Mitigation strategy
Alert fatigue from false or low-value alerts	4	5	**20**	Plausibility filtering, patient-specific thresholds, severity tiers, duplicate suppression windows, and alert-rate monitoring
Missed critical event (false negative)	2	5	10	Critical rules exempt from suppression, escalation timer, and scripted scenario tests for each critical rule
Breach or unauthorized access to health data	2	5	10	TLS, encryption at rest, role-based access, audit logging, and synthetic data only during development
Risk model inaccurate or biased	4	3	**12**	Baseline comparison, holdout validation, subgroup metrics by age and sex, and clinician review of all recommendations
Device nonuse or gaps in patient data	3	4	**12**	Offline buffering with retry, a missing-data pattern alert, and a minimal device set
Schedule overrun with one-week phases	3	4	**12**	MoSCoW scope control, parallel alert and dashboard work in Phase 4, and early deferral of Should items
Real-time performance degrades under load	3	3	9	Early load tests, indexes on patient and timestamp, and caching in Phase 6
Clinicians over-rely on automated recommendations	2	4	8	System-generated labels, visible contributing factors, and required clinician approval

Note. L = likelihood and I = impact, each rated 1 (low) to 5 (high); score = L × I. Scores of 12 or higher are shown in bold and are reviewed at every milestone.

Alert fatigue scores highest because it can make the system unsafe even when every component works as designed, so alert volume per patient will be tracked as a quality metric from Phase 4 onward. Model risk follows from the reliance on synthetic data: the risk module will be validated against a documented baseline and presented as decision support, with no claim of clinical accuracy beyond what synthetic validation supports.

Project Timeline and Milestones

Figure 2 presents the project timeline from October 19 to December 6, 2026. The work is divided into seven one-week phases: initiation and proposal, data layer and API design, backend implementation, the real-time alert system and monitoring dashboard, risk assessment and reporting, performance optimization, and final integration with delivery and presentation. Each phase ends in a milestone (M1 to M7) tied to its deliverables. Phase 4 carries the heaviest load because the alert system and dashboard are due together, so dashboard layout work begins against mock data while the alert rules are built, and the two converge once escalation logic is ready for real-time display. Two further dependencies shape the schedule: the synthetic data generator built in Phase 3 supplies training data for the predictive models in Phase 5, and the FastAPI endpoints must exist before alert rules can consume incoming readings. Should-priority requirements serve as the schedule buffer and are deferred first if a phase runs long.

References

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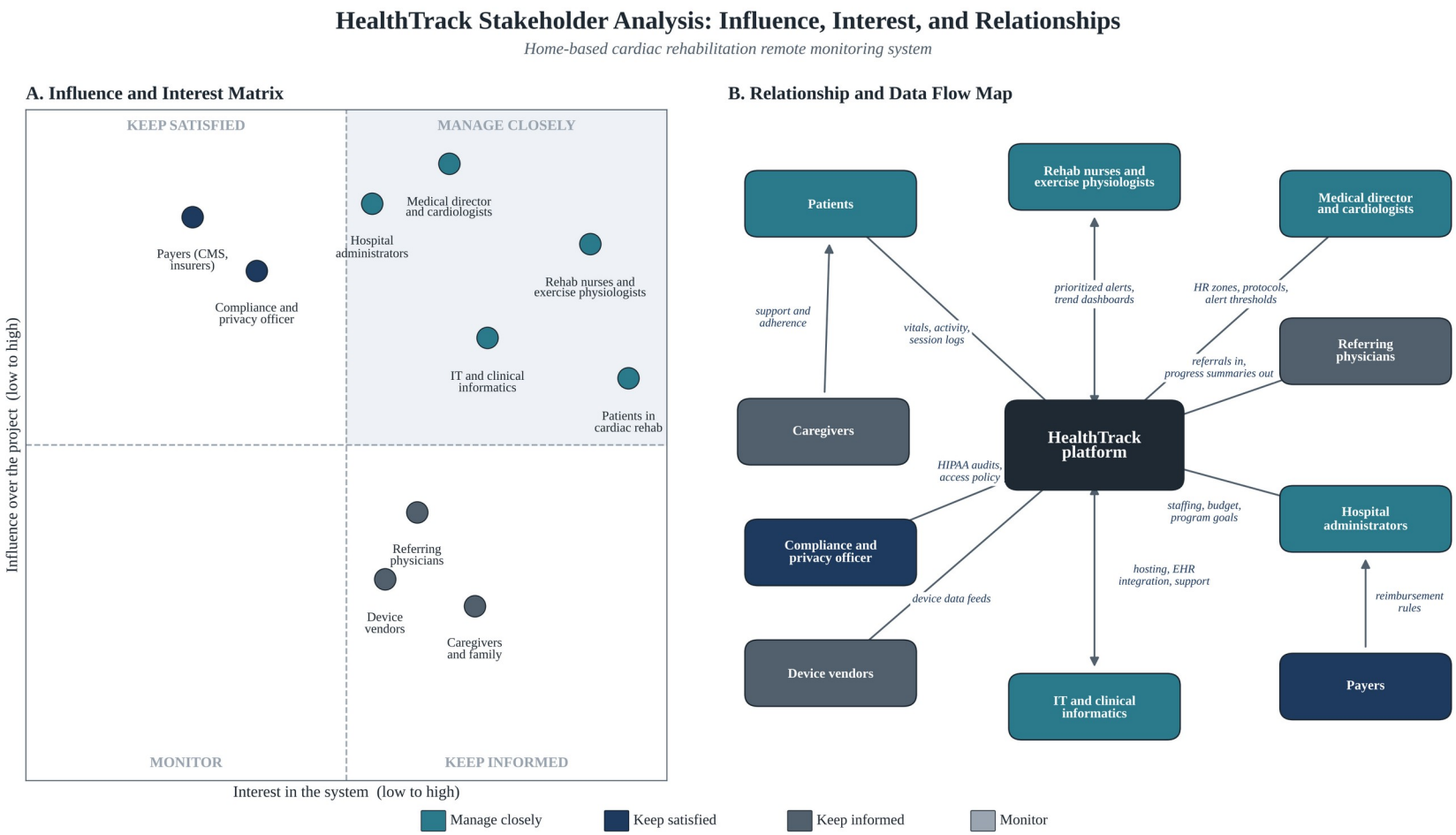
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Figure 1

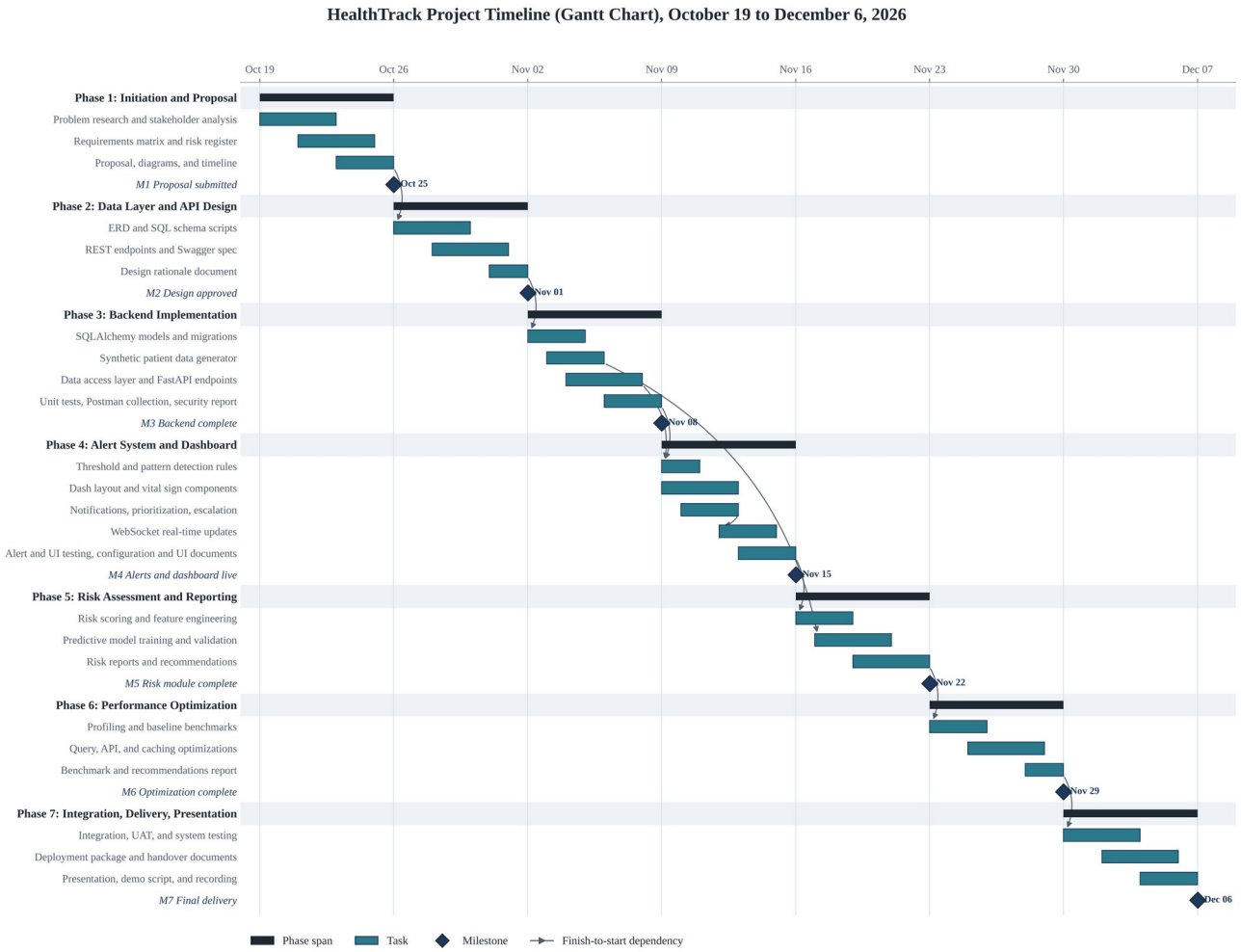
HealthTrack Stakeholder Influence and Interest Matrix and Relationship Map



Note. Panel A positions each stakeholder group by interest and influence. Panel B shows the data, authority, and funding flows among stakeholders and the platform. The full-size diagram is submitted separately.

Figure 2

HealthTrack Project Timeline, Phases, Milestones, and Dependencies



Note. Bars show task durations, diamonds mark phase milestones, and arrows show finish-to-start dependencies. The full-size chart is submitted separately.